

BDS DSC312 - Computer Vision with Multi-modal models & Analytics

Project Phase 2: Advanced Implementation, Methodology & Final Evaluation

Overview

This document outlines the requirements for Stage 2 (Final) of your computer vision project. While Stage 1 focused on feasibility and proof of concept, Stage 2 is where you move into high-performance engineering and academic contribution.

The objective of this stage is to move beyond a basic running pipeline and produce results that are comparable to, or exceed, the latest published results on your chosen dataset. You are no longer just using models; you are designing, modifying, and critically evaluating them to prove their efficacy and generalizability.

In this stage, I will discuss the basic requirements to "pass" this project and the advanced bonus requirements to achieve the [DICTA 2026](#) Challenge. If you achieve a DICTA-level publication, you get a full mark in this project. Are you ready?

Logistics

Total Marks: 40% of the unit's total.

Deliverables: (**one submission** only per team)

- Jupyter Notebook (.ipynb): Clean, documented, and reproducible.
If you are coding on Colab or Kaggle, provide a link that allows me to run your Notebook.
For the DICTA Challenge students: If your code is big, you may share your GitHub repo.
- Report:
 - Format: Word file.
 - Word count: 3000 words max (5-6 pages). **We will stop reading past the count.**
 - Use the provided IEEE Conference Template.
 - **For the DICTA Challenge students**: 4000-5000 words (8 pages max with references).
- Presentation:
 - Format: PowerPoint file.
 - Time restrictions: 10 minutes (usually, 10 slides maximum, but may vary).

Timeline:

- Submit report and code: Wed 1/4/2026 - 11:59 PM. (No late submissions).
 - Presentation: Thu 2/4/2026 in class: 9:30 AM to 1 PM according to a schedule I will publish later. Each group will present for 10 minutes, followed by 4-5 minutes of Q&A.
 - Submit the presentation after including my feedback: Thu 2/4/2026 - 11:59 PM.
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1. Defining Your Contribution

To achieve high marks, your project must demonstrate a contribution in one or more of the following categories:

1. **New Problem/Variation**: Solving a specific sub-problem or a unique variation of an existing task that has limited existing research. In other words, you are bridging the literature gaps.
2. **Architectural/Mathematical Innovation**: Implementing modifications to an existing architecture (e.g., adding an FPN map), designing a custom loss function or mathematical innovation, or combining CNN blocks or attention modules to a backbone in a novel way.

3. Superior Performance: Outperforming established baselines through superior engineering and design (not just hyperparameter tuning).

Note: Strong projects (DICTA-level) typically combine contributions 2 and 3 (combining 1 and 3 is also an option). Top-tier projects (better than DICTA) achieve all three. If you can only achieve contribution #3, you will get a good mark, but not the highest. If all you do is hyperparameter tuning to achieve good accuracy, you will still get a passing mark, provided all your other requirements are met according to the marking rubric.

2. Final Report Structure

Use the IEEE two-column template to maintain a professional standard. Your report should be structured as follows:

Abstract (150-250 words): A concise summary of the problem, your proposed method, and key results.

Section I: Introduction: A refined problem definition and the motivation behind your project, with key literature, highlighting their gaps, i.e., what is missing in the literature that you will achieve in your project?

Section II: Related Work (Extended): An expanded literature survey of what you did in Stage 1, which identifies the top datasets for your problem and establishes the current state-of-the-art (SOTA) performance. While the golden standard is recent literature from the last 5 years, you can include highly influential older papers.

For the DICTA-challenge students, an ideal paper should have 15-20 references (max 30).

Section III: Methodology: Detail your technical approach. Describe your architecture modifications or loss function designs in depth. Describe your modules or blocks and how you interconnected them. Diagrams are your best option here. I heard that [PaperPlot](#) does a good job, but I have not tried it myself, and it is not free!

Section IV: Experiments & Results: Compare your results directly against benchmarks published in the literature. I expect everyone to achieve results **comparable** to published ones. If your results are much lower, provide a convincing explanation of why this happened. Give a technical explanation. Don't just say that you ran out of time!

DICTA-challenge students:

- Achieve results **better** than state-of-the-art (SOTA) on published benchmarks
- Generalisability: Demonstrate performance on a **second dataset** to prove your model isn't overfitted to one specific environment.

Section V: Discussion of Results: What specific part of your design contributed most to the success? Perform an **ablation study**, e.g., turn off your custom loss or architecture block and measure the new performance to prove the value of your contribution.

Section VI: Conclusion: Final summary and potential future work.

Non-numbered Section: References

3. Presentation Outline (The "10-in-10" Rule):

To ensure we hear from all 12 groups within our 3-hour session, presentations are strictly limited to **10 minutes** followed by 5 minutes of Q&A. To hit this timing, your deck should not exceed **10-12 slides**. Below is one recommended slide distribution based on technical prominence:

#	Section	Content Focus	Timing
1	Title Slide	Project Title, Authors, and Affiliations (S P Jain).	30s
2	Outline	Roadmap of the presentation.	30s
3	Motivation	Why does this problem matter in the real world? (e.g., Mining safety, Sports analytics).	1m
4	Problem Definition	The specific CV challenge (Input/Output/Complexity).	1m
5	Literature Survey	1 Slide Only. Summary of the current body of work and the gaps you identified.	1m
6-7	Methodology	The Core. Diagram of your architecture or loss function modifications. Explain your innovation.	2m
8	Experiments	Dataset details and the SOTA Comparison table.	1.5m
9	Discussion	Visualising results (Success vs. Failure cases) and your Ablation Study.	1.5m
10	Conclusion	Summary of contributions and future outlook.	1m

Presentation Strategy for Success:

- **Visuals over Verbiage:** Use diagrams for architecture and heatmaps/plots for results. Avoid walls of text; if I am reading your slide, I am not listening to you.
- **The "Demo" Rule:** If you want to show a video of your model working (e.g., tracking or action recognition), embed it into your slides. Do not try to switch windows to a media player—it will eat your 10 minutes.
- **Hard Cut-off:** At the 10-minute mark, I will signal you to wrap up. Professionalism includes time management.
- **Zero Transition Time:** While Group N is answering questions, Group N+1 must be at the front, ready to plug in.

4. Marking Rubric (40 Marks)

Rubric	Marks	Criteria
Methodology and Innovation	10	Clear technical contribution. Depth of architectural or loss function modification (beyond simple hyperparameter tuning).
Results & Evaluation	10	Direct comparison against SOTA/published benchmarks. Proof of performance and/or generalisability across data.
Critical Analysis	6	Quality of the Ablation Study. Insightful discussion of failure cases and why the model behaves as it does.
Presentation	6	Effective communication of the problem and solution within 10 minutes. Professionalism in Q&A.
Code & Reproducibility	4	Clean, organized Notebook with Markdown headers. Results match report screenshots. Relative paths used.
Professionalism	4	Adherence to IEEE template and strict word counts. Quality of technical writing and figure clarity.

5. The DICTA 2026 Challenge

For teams that satisfy the advanced requirements, i.e., performing a thorough literature review, identifying two top datasets, and outperforming published results on both through significant design modifications, I will provide guidance on submitting your report as a formal paper to [DICTA 2026](#), the Australian flagship conference on computer vision.

Since all of you are graduating this semester (hopefully!), Stage 2 is more than a grade; it is a professional milestone. I want to encourage you to aim for the "Best Project" track, which isn't just about high marks; it's about positioning yourself for what comes next. If your results outperform published benchmarks or demonstrate exceptional design, your report can be adapted into a formal submission for DICTA 2026.

The 2026 conference is particularly special because it is being held right here in Sydney, offering a rare opportunity to network with the best in the field without leaving the city. Whether you are among the 90% heading straight into the Australian industry or are planning to pursue a Master's degree (here or in the US), a publication in an IEEE-sponsored conference like DICTA is your strongest professional currency. For industry-bound students, it proves you are a problem solver who can outperform established baselines, a skill highly valued by top-tier tech firms. Demonstrating that you can modify an architecture and perform a rigorous evaluation proves you have the technical depth that distinguishes a senior engineer from a junior who can only `pip install` an existing model. For those staying in academia, a peer-reviewed paper is the ultimate differentiator for securing competitive research scholarships and positions. This is your chance to graduate not just with a degree, but with a recognised contribution to the world of Computer Vision.

Good luck. It is time to see what your models are truly capable of!

- Hazem El-Alfy